

AFSHAL GULAM

afshal.gulam@u.nus.edu | linkedin.com/in/afshal-g | github.com/AfshalG

Availability: Aug 2026 – Jul 2027, San Francisco Bay Area | **Work Authorization:** Singapore Citizen, eligible for 12-month J-1 US visa, fully sponsored by NUS, hosted by Stanford University — zero cost or paperwork to the employer.

Computer Engineering student specializing in agentic AI systems and full-stack engineering. I focus on the human side of software — making complex AI feel intuitive — and move fast across the stack, validating problems before writing code and picking up new technologies from scratch when the problem demands it.

EDUCATION

Bachelor of Engineering, Computer Engineering Aug 2024 – Jun 2028
National University of Singapore (NUS)

EXPERIENCE

Phase 2, TinyFish Accelerator Feb 2026 – Present
Selected as 1 of 10 builders from 1,500+ applications | \$2M seed pool, backed by Mango Capital

Lead Residential Assistant, NUS UTown Residence May 2025 – May 2026
National University of Singapore

- Selected as Lead RA for one of six residential neighbourhoods; organised and led campus-wide event for 1,000+ residents across UTown Residence.

PROJECTS

Solo Founder, AgentDesk – AI Agent Verification Desktop App Feb 2026 – Present
TinyFish Accelerator Phase 2 | Announced by TinyFish

- Built a native macOS app in Rust + Tauri v2 with async tokio backend, hash-based port allocation, and HTTP hook receiver for Claude Code sessions; 4 public releases shipped in 2 months.
- Integrated production stack: TinyFish for real-browser E2E via SSE, Fireworks AI for 1.4s failure diagnosis (gpt-oss-120b) and RAG-based Suggest Fix (Kimi K2.5), Composio for GitHub PR and Slack workflow automation.
- Engineered verification pipeline driving real browsers through production OAuth, deploy, and E2E flows; secrets firewall blocks agent reads of .env, SSH keys, and API key patterns.
- Embedded Claude Code as in-app subprocess for built-in agent chat with persistent transcripts, message rewind, and TCC-aware sandbox handling on unsigned builds.

Co-Founder, TaskGenie – Agentic AI Task Management Platform May 2025 – Dec 2025
Featured by Google | NUS Orbital Judges' Choice Award (out of 1,000+ teams)

- Architected multi-agent AI system with 16 function-calling tools via Vercel AI SDK and LangGraph, reducing manual operations by 85%.
- Implemented advanced vector search using 2000-dimensional Gemini embeddings with DPG diversification, enabling semantic task discovery across 10+ database tables with sub-500ms query times.
- Developed production-scale Next.js 15 application with server actions, type-safe mutations, and SWR hooks for optimistic UI updates across 155+ custom components.
- Achieved 70%+ test coverage using Vitest and React Testing Library; enforced RLS, CSP, OAuth 2.0, and intelligent caching strategies with 20ms streaming architecture.

Technical Lead, Smart Rehab Band – IoT Healthcare Innovation Aug 2025 – Oct 2025
NUS IDEATE 2025 – Semifinalist (Top 16 of 500+ teams)

- Engineered IMU-based wearable providing real-time haptic coaching for range of motion, cadence, and smoothness — privacy-first alternative to video-based rehabilitation tools.
- Secured \$1,000 prototyping fund and NOC Greenlane access; led four-person team to semifinals with sub-\$70 MVP built on Arduino Nano 33 BLE.

Multi-Domain AI System – Computer Vision, RL, Document Reconstruction May 2025 – Jun 2025
DSTA BrainHack AI/ML Engineering Competition – Semifinalist

- Engineered YOLOv8m object detection pipeline with custom preprocessing and hyperparameter optimization, achieving 52.1% mAP across 18 vehicle classes; implemented PPO reinforcement learning agent achieving 99.4% efficiency over 750K timesteps.
- Developed fragment-based document reassembly algorithm (77.6% accuracy) within 24-hour constraint; containerized 3 models with Docker and FastAPI on GCP, maintaining subsecond inference and 100% uptime across 1,400+ test cases.

SKILLS

- Languages** – TypeScript, JavaScript, Python, Rust, Java, C, C++, SQL
- Frameworks** – Next.js, React, Node.js, Tauri v2, tokio, Anthropic SDK, LangGraph, Vercel AI SDK, PyTorch, FastAPI, Docker, Google Cloud Platform, Supabase
- Interests** – Agentic AI Systems, Developer Tooling, Human-AI Interaction, Full-Stack Product Engineering, Assistive HealthTech